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import os
import sys
import time
import datetime
import webbrowser
import speech_recognition as sr
import pyttsx3

# Initialize the text-to-speech engine
engine = pyttsx3.init()
voices = engine.getProperty('voices')
# Set voice to male (usually index 0, change to 1 for female depending
on your OS)
engine.setProperty('voice', voices[0].id)
engine.setProperty('rate', 175) # Set speaking speed

def speak(text):
    """Make the assistant speak the given text."""
    print(f"Jarvis: {text}")
    engine.say(text)
    engine.runAndWait()

def wish_me():
    """Greet the user based on the current time of day."""
    hour = int(datetime.datetime.now().hour)
    if 0 <= hour < 12:
        speak("Good Morning, sir.")
    elif 12 <= hour < 18:
        speak("Good Afternoon, sir.")
    else:
        speak("Good Evening, sir.")
    speak("I am Jarvis. How can I assist you today?")

def take_command():
    """Listen for microphone input and return it as a string
    string."""
    recognizer = sr.Recognizer()
    with sr.Microphone() as source:
        print("\nListening...")
        recognizer.pause_threshold = 1 # Seconds of non-speaking
audio before phrase is considered complete
        try:
            audio = recognizer.listen(source, timeout=5,
phrase_time_limit=8)
        except Exception:
            return "none"

    try:

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        print("Recognizing...")
        query = recognizer.recognize_google(audio, language='en-in')
        print(f"User said: {query}\n")
except Exception:
    # Return "none" if voice is not recognized clearly
    return "none"

return query.lower()

def execute_assistant():
    """Main logic loop for the assistant execution."""
    wish_me()

    while True:
        query = take_command()

        # Exit condition
        if "exit" in query or "quit" in query or "stop" in query or
"goodbye" in query:
            speak("Goodbye, sir. Have a great day.")
            sys.exit()

        # Open web pages
        elif "open google" in query:
            speak("Opening Google, sir.")
            webbrowser.open("https://www.google.com")

        elif "open youtube" in query:
            speak("Opening YouTube, sir.")
            webbrowser.open("https://www.youtube.com")

        # Tell the current time
        elif "the time" in query or "current time" in query:
            str_time = datetime.datetime.now().strftime("%I:%M %p")
            speak(f"Sir, the time is {str_time}")

        # Basic identity responses
        elif "who are you" in query or "what is your name" in query:
            speak("I am Jarvis, your virtual desktop assistant.")

        elif "how are you" in query:
            speak("I am functioning perfectly, sir. Thank you for
asking.")

        # Search query on Google
        elif "search" in query:
            search_query = query.replace("search", "").strip()
            if search_query:

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        speak(f"Searching Google for {search_query}")

webbrowser.open(f"https://www.google.com/search?q={search_query}")
    else:
        speak("What would you like me to search for, sir?")

    # Fallback for unrecognized phrases that successfully
    converted to string text
    elif query != "none":
        speak("I heard you, but I do not have a built-in command
for that action yet.")

if __name__ == "__main__":
    # Note: Requires 'pip install speechrecognition pyttsx3' and a
    working microphone
    execute_assistant()
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